Supplementary — The Ribbon at the Bell Bound

Supplementary: program history and infrastructure

Supplementary to Section 5 (Methods) of the main text. Moved here per the structural revision: this is phase chronology and implementation detail, not method.

SI-P.1 The four phases

Phase A built the computational scaffold — pure JAX (jit, vmap, lax.scan; no Python loops over steps or batch), float32 state as explicit pytrees — and passed the R0 null control, the baseline that later grew into the regression suite carried through phase C (commit 86e2154, “phase-A: skeleton+tests+R0 green”). It fixed no physics on its own; it fixed the ground on which every later claim would have to stand.

Phase B established gauge blindness. Any observable that is a function of the boundary axes $n = R(q)\hat{e}$ factors through the covering map $SU(2) \to SO(3)$ and is therefore identically blind to the $\mathbb{Z}_2$ lift; this is a theorem (Section 5), first met here as an empirical wall. The operative number was energetic: kinks carry energy of order $k\_e \cdot (\pi/2..\pi)^2$ and remain thermally unpopulated for $T \ll k\_e$, so the charged sector never populates spontaneously (ribbon_model_note.md § “Дополнение (июль 2026, после фазы B симуляции): no-go гейдж-слепоты и роль вложения в 4D”).

Phase C produced the entropic singlet. With a conserved $\mathbb{Z}_2$ charge (the 2π twist sector) read under an entropic basin measure, the correlation takes the singlet form $E(\theta \mid 2\pi) = -\tanh(\Delta S \cdot \cos\theta \;/\; 2)$ with $\Delta S \propto \cos\theta$ (ribbon_model_note.md § “Резолюция открытой задачи §3 (фаза C симуляции, июль 2026)”; phase_C_report). The final phase-C lemma fixed its ceiling: the amplitude $\tanh(\Delta S/2) \approx 0.5$ stays strictly below 1, so exact zeros — and with them the Born rule — are unreachable by a classical entropic measure. The ceiling was double: the phase-C lemma also recorded that the effect was finite-size — the 2π charge is carried by a twist soliton, localized (peak twist density ≈ 0.9), whose effect passes through a crossover in chain length — ferromagnetic sign up to $N-1 \approx 33$, an antiferromagnetic window near 48–64, extinction by $N-1 \approx 95$ — so the singlet form is finite-size and fades as $N \to \infty$ (phase_C_report §7). Exact zeros and scale invariance are both defining properties of the quantum correlation; the classical entropic mechanism had neither. These two missing invariances are precisely the questions phase D inherited. (The CHSH figure quoted at the time was computed with an estimator we later withdrew; see Section 7.)

Phase D embedded the framed ribbon in $R^4 \equiv \mathbb{H}$, turning the integer twist ($\mathbb{Z}$) into a $\mathbb{Z}_2$ framing class ($\pi_1(SO(3)) = \mathbb{Z}_2$). Across the D0–DS3 sequence — infrastructure and invariant (D0), the readout operator and empty-sector test (D1), scaling (D2/D2-ext), amplitude origin (S1), cross-scan and CHSH audits (DS2/DS3) — the single number that survived is the length-stable readout amplitude (no resolved decay over the measured range) A_plateau = 0.363 ± 0.012 (D2-ext; commit 2784edf), whose meaning and limits are the subject of Section 6.

SI-P.2 Infrastructure and test coverage

Phase D reused the phase A–C toolkit directly: the quaternion frame algebra, energy, and Langevin dynamics carried over into the R^4 model (band4d.py, measurement.py). Tests were organized by INVARIANT rather than by count: Bishop SO(4) transport ($R \cdot t = t'$, $R^T R = I$, det $= +1$, identity on the orthogonal complement — minimality), the $\mathbb{Z}_2$ parity invariant (T-inv-1/2/3: straight rod → +1, inserted 2π → −1, 4π → +1, and invariance under 1000 smooth deformations), mirror symmetry, and singularity detection (temporal lift jump, tangent reversal, spatial lift-wall). The phase-D suite (9 tests) and the phase A–C suite (60 tests) both run green under this coverage. Phases A–C run in float32 per the simulation defaults (SPEC §5); all phase-D runs are CPU-JAX in float64. We chose float64 for the lift bookkeeping: parity is the sign of a quaternion lift accumulated along the whole chain (invariant.py), and we judged that a sign read off an accumulated product, together with its conservation test (T-inv-3), wanted precision headroom we did not want to argue about. The runs are CPU-only because fp64 on the available GPU is ~1/64 speed, and cells this size do not need it.

Supplementary: AI-assisted methodology (detailed)

Supplementary to Section 9.2 of the main text. Moved here per journal-policy formatting of the AI disclosure; the main text carries the disclosure statement, this file carries the working arrangement and its evidence.

SI-M.1 The three-party arrangement and two-directional self-correction

The program was run by three parties: an AI architect (Claude — design, pre-registration, review), an AI executor (Claude Code — implementation, runs, and verification of every drafted claim against its cited source), and a human owner (decisions and veto). The controls of Section 4.1 are controls over both AI roles, not over the code alone: pre-registration with kill-criteria and the raw-before-analysis order bind design and execution to commitments made before the data existed, which is precisely where a shared preference for a positive result would otherwise enter. Layered on top, each side checks the other against primary sources — errors were caught in both directions, and the direction of capture is documented in the commit history. In the architect's direction: a misattributed number in a discussion draft — a value belonging to an analytical construction, quoted as though measured — was caught by the executor's check against the canonical record and never entered the text. In the executor's direction: a fabricated test count was caught by the executor's own check against the test runner, and a reproducibility alarm the executor raised on three points was refuted by the executor's own pre-registered eleven-point audit, which found the quoted errors honest (Section 6.1). Neither role is reliable unaudited, and neither audits itself by inspection; what makes the arrangement work is that the commitments are written down first and the checks run against sources rather than against memory.